

Erica Lindsey Busch

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Academic appointments

College of Connected Computing, Vanderbilt University
Incoming Assistant Professor, Computational Psychology Area

upcoming: AY 2027/2028 –
Nashville, TN, USA

Center for Neuroscience Imaging Research, Sungkyunkwan University
Institute for Basic Science Young Scientist Fellow/Research Professor

August 2026 – Present
Suwon-si, South Korea

Education

Yale University
Ph.D. with Distinction, Neuroscience/Psychology (M.S. 2022, M. Phil. 2023)
Thesis: Understanding and enhancing human cognition along neural manifolds
Advisor: Nicholas Turk-Browne

August 2020 – May 2026
New Haven, CT, USA

Dartmouth College
B.A. with High Honors in Cognitive Science, Computer Science
Thesis: A deep learning approach to scene perception in autism
Advisors: James Haxby, Caroline Robertson

September 2016 – March 2020
Hanover, NH, USA

Centro Tinku Academic Institution
Advanced Language Study Abroad in Spanish

August – November 2017
Cusco, Peru

Awards and honors

Institute for Basic Science Young Scientist Fellowship (300 million KRW annually)
Project title: Computational modeling of affective brain-state trajectories for personalized neurofeedback

2026–2029

Conference Travel Fellowship (\$800 awarded twice)
Yale University Graduate Student Assembly

2025 & 2026

Doctoral Dissertation Distinction Award
Degree Committee, Yale University Graduate School of Arts and Sciences

2026

Society for Neuroscience Annual Meeting Travel Award (\$500 awarded twice)
Wu Tsai Institute at Yale University

2023 & 2025

ReproNim/INCF Fellowship (\$700)
Center for Reproducible Neuroimaging Train-the-Trainer Program

2024–2026

Google PhD Fellowship (Finalist)

2024

Social and Affective Neuroscience Society Naturalistic Data Challenge (First Prize)
Project title: Relating neural dynamics and emotion dynamics with nonlinear manifold learning

2022

National Science Foundation Graduate Research Fellowship (\$138,000)
Project title: Enhancing human learning along neural manifolds

2021–2024

Outstanding Undergraduate Research Award (Second Prize)
Neukom Institute for Computational Science

2020

“Made at Dartmouth” Research Competition Winner (First Prize)
Dartmouth College Undergraduate Advising and Research, [video](#)

2020

Academic Achievement Prize in Cognitive Science

2020

Fulbright Fellowship (Withdrew due to COVID-19)	2020
High Honors in Cognitive Science	2020
William H. Neukom Scholarship (\$1,000 awarded twice) <i>Neukom Institute for Computational Science</i>	2019-2020
National Science Foundation Research Experience for Undergraduates (\$5,500) <i>University of Washington Center for Neurotechnology</i> <i>Project title: Deep learning models of octopus decision-making</i>	2019
David C. Hodgson Endowment Award for Cognitive Neuroscience Research (\$5,000) <i>Dartmouth College Undergraduate Advising and Research</i>	2019
James O. Freedman Presidential Scholarship (\$1,000 awarded twice) <i>Dartmouth College Undergraduate Advising and Research</i>	2018-2019
Sophomore Research Scholarship, Dartmouth College (\$1,000) <i>Dartmouth College Undergraduate Advising and Research</i>	2018
Honors List, Dartmouth College	2017-2020
National Merit Scholarship (Finalist)	2015

****** Awarded scholarship, fellowship, & grant amounts in USD unless noted; only select awards included

Publications

Peer-reviewed articles and conference proceedings

Busch, E.L., Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2026). Human learning of noninvasive brain-computer interfaces via manifold geometry. *Nature Neuroscience*. doi:10.1038/s41593-026-02311-2

- Analysis repository: github.com/ericabussh/avaterRT_analysis
- Task repository: github.com/ericabussh/avaterRT_analysis
- Curated dataset: doi.org/10.5061/dryad.9cnp5hr0w

Busch, E.L., Turk-Browne, N.B., & Baskin-Sommers, A.R. (2026). Revamping neuroimaging analysis to reveal biomarkers of adolescent mental health. *Nature Mental Health*. 4, 486-498. doi:10.1038/s44220-026-00610-y

- Tutorial and data repository: github.com/ericabussh/RepBiomrkr

Busch, E.L. & Turk-Browne, N.B. (2025). Intrinsic dimensionality of brain activity manifolds across tasks and development. *Proceedings of the 8th Annual Conference on Cognitive Computational Neuroscience*.

Afrasiyabi, A., Bhaskar, D., **Busch, E.L.**, Caplette, L., Singh, R., Lajoie, G., Turk-Browne, N.B., & Krishnaswamy, S. (2025). Latent representation learning for multimodal brain activity translation. *IEEE International Conference on Acoustics, Speech, and Signal Processing [ICASSP2025]*. 10.1109/ICASSP49660.2025.10887834

Busch, E.L.*, Conley, M.I.*, & Baskin-Sommers, A. (2024). Manifold learning uncovers nonlinear interactions between the adolescent brain and environment that predict emotional and behavioral problems. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*. doi.org/10.1016/j.bpsc.2024.07.001

- Analysis repository: github.com/ericabussh/manifold_abcd_psychopathology_bpcnni
- PIP package: pypi.org/project/EPHATE

Roskies, A., **Busch, E.L.**, & Walton, A. Agency as a framework for thinking about neuropsychiatric disease: A prelude to asking causal questions. *Causal Concepts in Psychopathology: Multidisciplinary Perspectives*, Cambridge University Press.

Busch, E.L., Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2024). Learning along the manifold of human brain activity via real-time neurofeedback. *Proceedings of the 7th Annual Conference on Cognitive Computational Neuroscience*.

Busch, E.L., Rapuano, K.M., Anderson, K.M., Rosenberg, M.D., Watts, R., Casey, B.J., Haxby, J.V., & Feilong, M. (2024). Dissociation of reliability, predictability, and heritability in fine- and coarse-scale functional connectomes during development. *Journal of Neuroscience*. 44(6), doi:10.1523/JNEUROSCI.0735-23.2023.

- Analysis repository: github.com/ericabussh/ABCD_hyperalignment_JNeurosci

Skalaban, L.J., Chan, I., Lin, Q., Rapuano, K.M., Conley, M.I., **Busch, E.L.**, Watts, R., Murty, V., & Casey, B.J. (2024). Representational dissimilarity of faces and places during a working memory task is associated with subsequent recognition memory during development. *Journal of Cognitive Neuroscience*. 36(3) 415-434, doi:10.1162/jocn.a.02094.

Afrasiyabi, A., **Busch, E.L.**, Singh, R., Bhaskar, D., Capette, L., Turk-Browne, N.B., Krishnaswamy, S. (2024). Looking through the mind's eye via multimodal encoder-decoder networks. *Machine as Medium: Proceedings of the Center for Collaborative Arts and Media*, Fall 2024 Volume. doi:10.48550/arXiv.2410.00047.

Busch, E.L., Yates, T.S., & Turk-Browne, N.B. (2023). Tasks collapse the intrinsic dimensionality of activity in non-selective cortex. *Proceedings of the 6th Annual Conference on Cognitive Computational Neuroscience*.

Busch, E.L., Huang, J., Benz, A., Wallenstein, T., Lajoie, G., Wolf, G., Krishnaswamy, S., & Turk-Browne, N.B.* (2023). Multi-view manifold learning of human brain-state trajectories. *Nature Computational Science*. 3(3), 240-253, doi:10.1038/s43588-023-00419-0

- Analysis repository: github.com/ericabusht/tphate_analysis_capsule
- PIP package: pypi.org/project/TPHATE

Busch, E.L. & Krishnaswamy, S. (2023). Revealing trajectories of the mind via non-linear manifolds of brain activity. *Nature Computational Science*. 3(3), 192-193, doi.org/10.1038/s43588-023-00423-4 *Invited research briefing*.

Huang, J*, **Busch, E.L.***, Wallenstein, T*, Gerasimiuk, M., Benz, A., Lajoie, G., Wolf, G., Turk-Browne, N.B., & Krishnaswamy, S. (2022). Learning shared neural manifolds from multi-subject fMRI data. *Proceedings of the 32nd IEEE Machine Learning for Signal Processing*. doi:10.1109/MLSP55214.2022.9943383

Busch, E.L.*, Slipski, L*, Feilong, M., Guntupalli, J., Visconti di Oleggio Castello, M., Huckins, J., Nastase, S., Gobbini, M.I., Wager, T., & Haxby, J. (2021). Hybrid hyperalignment: A single high-dimensional model of shared information embedded in cortical patterns of response and functional connectivity. *NeuroImage*. 233, doi:10.1016/j.neuroimage.2021.117975

- Analysis and software repository: github.com/ericabusht/hybrid_hyperalignment_neuroimage

Manuscripts

Busch, E.L., & Turk-Browne, N.B. Developmental tuning of functional manifold dimensionality across the human brain. (Submitted)

*Denotes equal contribution.

Invited talks

Symposium on Closed-Loop Neurofeedback <i>Organization for Human Brain Mapping Annual Meeting, Bordeaux, France</i>	June 2026
Seminar in Computational Psychology <i>College of Connected Computing, Vanderbilt University, Nashville, TN</i>	Feb. 2026
NYU Concats <i>Psychology Department, New York University, New York, NY</i>	Jan. 2026
Seminar in Computational Psychology <i>Department of Psychology, Tufts University, Boston, MA</i>	Dec. 2025
Cognition, Perception, & Cognitive Neuroscience Seminar Series <i>Department of Psychological and Brain Science, University of California - Santa Barbara, Santa Barbara, CA</i>	Nov. 2025
Machine Learning Seminar Series <i>Department of Systems & Computational Biology, Albert Einstein College of Medicine, New York, NY</i>	Nov. 2025
Workshop on Functional Alignment of Information Encoded in Neural Activity <i>Center for Cognitive Neuroscience, Dartmouth College, Hanover, NH</i>	Sept. 2025
Center for Neuroscience Imaging Research Seminar <i>Institute for Basic Science, Suwon, South Korea</i>	July 2025
International Symposium on Decoded Neurofeedback <i>Advanced Telecommunications Research Institute, Kyoto, Japan</i>	July 2025

Manhattan Area Memory Meeting <i>Columbia University's Zuckerman Institute, New York, NY</i>	June 2025
fMRI Brown Bag <i>Center for Cognitive Neuroscience, Dartmouth College, Hanover, NH</i>	May 2025
CompCog Joint Lab Meeting <i>Department of Cognitive Science, Yale University, New Haven, CT</i>	Apr. 2025
Schoenbaum and Kahnt Labs Meeting <i>National Institute on Drug Abuse, Baltimore, MD</i>	Apr. 2025
Cognitive Science Group Meeting <i>Department of Cognitive Science, Johns Hopkins University, Baltimore, MD</i>	Apr. 2025
Magnetic Resonance Research Center fMRI Lecture Series <i>Department of Radiology, Yale University, New Haven, CT</i>	Dec. 2024
Kavli at Yale 20th Anniversary Symposium <i>Kavli Institute for Neuroscience, Yale University, New Haven, CT</i>	Sept. 2024
ABCD Insights & Innovations Meeting <i>National Institute of Health, Bethesda, MD</i>	Mar. 2024
BrainWorks Projects in Progress <i>Wu Tsai Institute at Yale University, New Haven, CT</i>	Nov. 2023
Shine Lab Meeting <i>The University of Sydney, (Virtual)</i>	Apr. 2023
MRI Users Meeting <i>FAS Brain Imaging Center, Yale University, New Haven, CT</i>	Oct. 2022
Imaging Analytics Meeting <i>Adolescent Brain and Cognitive Development (ABCD) project resting-state fMRI working group (Virtual)</i>	Sept. 2022
Current Works in Behavior, Genetics, and Neuroscience, Yale University <i>Yale University Department of Psychology, New Haven, CT</i>	Apr. 2022
FINN Lab Meeting <i>Department of Psychological and Brain Sciences, Dartmouth College (Virtual)</i>	Apr. 2021

Conference presentations

Busch, E.L., Krishnaswamy, S., & Turk-Browne, N.B. (2025) Neural dimensionality expands over the course of brain-computer interface learning. *Poster at the Society for Neuroscience Annual Meeting*. San Diego, CA, USA.

Busch, E.L., & Turk-Browne, N.B. (2025) Intrinsic dimensionality of brain activity manifolds across tasks and development. *Poster at 8th Annual Conference on Cognitive Computational Neuroscience*. Amsterdam, Netherlands

Busch, E.L., & Turk-Browne, N.B. (2025) Developmental differences in the intrinsic dimensionality of regional brain activity. *Poster at the Cognitive Neuroscience Society Annual Meeting*. Boston, M.A., USA.

Busch, E.L., Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2024) Learning along the manifold of human brain activity via real-time neurofeedback. *Oral Presentation at the Real-time Functional Imaging and Neurofeedback (rt-FIN) Meeting*. Heidelberg, Germany.

Busch, E.L., Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2024) Learning along the manifold of human brain activity via real-time neurofeedback. *Contributed Talk and Poster at 7th Annual Conference on Cognitive Computational Neuroscience*. Cambridge, M.A., USA.

Busch, E.L., Conley, M.I., & Baskin-Sommers, A. (2024). Using manifold learning to uncover the embedded brain and implications for mental health in youth. *Poster at the Organization for Human Brain Mapping Annual Meeting*. Seoul, South Korea.

- Busch, E.L.**, Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2024). Learning on the manifold of human brain activity through real-time neurofeedback. *Poster at the Organization for Human Brain Mapping Annual Meeting*. Seoul, South Korea.
- Busch, E.L.**, Fincke, E.C., Lajoie, G., Krishnaswamy, S., & Turk-Browne, N.B. (2023). Learning on the manifold of human brain activity through real-time neurofeedback. *Talk at the Society for Neuroscience Annual Meeting Nanosymposium on Neural Decoding and Neuroprosthetics*. Washington, D.C., USA.
- Busch, E.L.**, Yates, T.S., & Turk-Browne, N.B. (2023). Tasks constrain the intrinsic dimensionality of activity in non-selective cortex. *Poster at the 7th Annual Conference on Cognitive Computational Neuroscience*, Oxford, United Kingdom.
- Busch, E.L.**, Bhaskar, D., Letrou, A., Zhang, X., Noah, J.A., Lajoie, G., Hirsch, J., Turk-Browne, N.B., Krishnaswamy, S. (2022). An encoder-decoder framework for cross-modal translation of brain imaging data. *Poster and selected lightning talk at the Montreal AI-Neuroscience Meeting*. Montreal, QC, Canada.
- Busch, E.L.**, Letrou, A., Huang, J., Lajoie, G., Wolf, G., Krishnaswamy, S., & Turk-Browne, N.B. (2022). A neural manifold learning framework for real-time fMRI neurofeedback. *Poster at the Society for Neuroscience Annual Meeting*. San Diego, CA, USA.
- Busch, E.L.**, Letrou, A., Huang, J., Lajoie, G., Wolf, G., Krishnaswamy, S., & Turk-Browne, N.B. (2022). A neural manifold learning framework for real-time fMRI neurofeedback. *Poster at the Real-time Functional Imaging and Neurofeedback Meeting*. New Haven, CT, USA.
- Busch, E.L.**, Rapuano, K.M., Anderson, K.M., Rosenberg, M.D., Watts, R., Casey, B.J., Haxby, J.V., & Feilong, M. (2022). Heritable template underlies reliable idiosyncrasies in the developing fine-scale connectome. *Poster at the Organization for Human Brain Mapping Annual Meeting*. Glasgow, Scotland.
- Letrou, A., **Busch, E.L.**, & Turk-Browne, N.B., (2022). Relating neural dynamics and emotion dynamics with nonlinear manifold learning. *Poster and talk at the Social and Affective Neuroscience Society Annual Meeting*.
- Roskies, A., Walton, A., Roth, R.M., **Busch, E.L.**, Holtzheimer, P.E., (2022). Measuring the dimensions of agency: A data-driven approach. *Poster at the Philosophy of Science Association*. Pittsburgh, PA.
- Busch, E.L.**, Huang, J., Benz, A., Wallenstein, T., Lajoie, G., Wolf, G., Krishnaswamy, S., & Turk-Browne, N.B. (2021). Manifold learning to capture brain-state trajectories in fMRI. *Poster at the Society for Neuroscience Annual Meeting*.
- Walton, A.E., Nizzi, M.C., West, B., Mofe, E., Roth, R.M., **Busch, E.L.**, Holtzheimer, P.E., & Roskies A.L. (2021). The impact of anxiety and depression on dimensions of agency. *Poster at the 7th Annual NIH BRAIN Initiative Annual Meeting*.
- Sivitilli, D.M., Weertman, W.L., **Busch, E.L.**, Ullmann, J.F., Smith, J.R., Gire, D.H. (2021). Strategies of single arm foraging in Octopus rubescens in the absence of visual feedback. *Poster at the Society for Integrative and Comparative Biology*.
- Busch, E.L.**, Haskins, A.J., Isik, L., & Robertson, C.E. (2020) A deep learning approach to understanding real-world scene perception in autism. *Presidential Undergraduate Research Symposium, Dartmouth College*.
- Walton, A.E., **Busch, E.L.**, Ratoff, W., Smith, W., Holtzheimer, P.E., & Roskies, A.L. (2020). Developing an agency assessment tool for understanding changes in agency with neurointerventions: Preliminary results. *Poster at the 6th Annual NIH BRAIN Initiative Annual Meeting*.
- Botch, T.L., **Busch, E.L.**, & Robertson, C.E. (2020). Application of deep neural networks to model omnidirectional gaze behavior in VR. *Poster at the Vision Sciences Society Annual Meeting*.
- Busch, E.L.**, Sivitilli, D.M., & Gire, D.H. (2019). Using deep learning to model octopus arm motion. *Poster and talk at the Center for Neurotechnology Research Symposium*. Seattle, WA, USA.
- Busch, E.L.**, Ma, F., Nastase, S.A., & Haxby, J.V. (2019). Individual differences in fine-grained neural correlates of mental states. *Poster at the Wetterhahn Science Symposium*. Hanover, NH, USA.

Teaching experience

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| Guest Lecturer, NEU 480: fMRI Decoding: Reading minds using brain scans, <i>Princeton University</i> | Feb. 2026 |
| Instructor, Interdepartmental Neurosci. Program, <i>Yale University</i> | Aug. 2024 & 2025 |
| fMRI Tutorial, INP First-Year Student Bootcamp. | |
| Guest Lecturer, CPSC 663: Deep learning, <i>Yale University</i> | Apr. 2025 |

Guest Lecturer, PSYC 220: Images of mind, <i>University of Illinois</i>	Mar. 2025
Teaching Fellow, Department of Psychology, <i>Yale University</i> PSYC 258/558/NCSC 258: Computational methods in human neuroscience.	Spring 2022 & 2023
Teaching Fellow, Department of Psychology, <i>Yale University</i> NCSC 160/PSYC 160: <i>The human brain</i> .	Fall 2022
Guest Lecturer, NCSC 270: Research methods in cognitive neuroscience, <i>Yale University</i>	Nov. 2021
Teaching Fellow, Department of Psychology, <i>Yale University</i> PSYC 270/NCSC 270: Research methods in cognitive neuroscience.	Fall 2021
Teaching Assistant, Department of Computer Science, <i>Dartmouth College</i> COSC 74: Machine learning and statistical data analysis	Spring 2020
Teaching Assistant, Department of Psychological & Brain Sciences, <i>Dartmouth College</i> PSYC 6: Introduction to neuroscience	Winter 2019
Peer Tutor, Tutor Clearinghouse, <i>Dartmouth College</i> SPAN 1-3 (Intro Spanish), SPAN 9 (Culture and Conversation: Advanced), SPAN 20 (Writing and Reading), COSC 1 (Intro to Programming and Computation), COSC 10 (Object-Oriented Programming), COSC 50 (Software Design), COSC 74 (Machine Learning), PSYC 6 (Intro to Neuroscience), PSYC 10 (Statistics), COGS 1 (Intro to Cognitive Science)	2017 - 2020
Cryptography Instructor, <i>Sonia Kovalevsky Math Day at Dartmouth College</i>	Apr. 2018
College Access Coach, <i>Let's Get Ready</i> Taught SAT and college application courses for students from historically underrepresented backgrounds in NYC public schools.	Summer 2017
Instructor, <i>Center for Gifted Youth, Long Island University</i> Designed and taught courses on mathematics and genetics for advanced middle school students.	Summer 2017
Private tutor K-12: NY State Regents math and sciences, English, History, Spanish; AP: Calculus AB and BC, Statistics, Physics, Computer Science; SAT / ACT; UG: Algebra, Graph Theory, Spanish.	2012–2024

Service and outreach

Rapporteur, CIFAR Brain, Mind, & Consciousness Program	2024-Present
Trainee Committee, Cognitive Computational Neuroscience Conference	2024
Organizer & Founding Member, Innovators in Cognitive Neuroscience Seminar Series	2020–Present
Fellow & Founding Member, Wu Tsai Institute Student, Postbac, & Postdoc Collective	2022–Present
Yale Psychology Colloquium Committee	2021–2023
Sneak Peek Mentor, Yale Psychology Diversity Committee	2021–2023
Facilitator and mentor, Dartmouth Leadership, Attitudes, and Behaviors Program <i>Ran 10-week workshops on value-driven leadership through the Nelson A. Rockefeller Center for Public Policy</i>	2018–2020
SIBS Mentorship Program Director, Dartmouth Center for Social Impact <i>Mentored for and directed a one-on-one mentorship program for Dartmouth undergrads and local youth involved in county social services. Coordinated with mentors, teachers, parents, and social workers; responsible for training mentors.</i>	2016–2020

Mentorship

Dominic Gearing (Yale undergraduate and honors thesis student) <i>Now: Associate at Kane & Company</i>	2024–2026
David Lee (Yale undergraduate) <i>Now: Healthcare Administrative Intern, Mayo Clinic</i>	2024–2026
E. Chandra Fincke (Yale undergraduate and honors thesis student)	2022–2024

Now: Master's student in AI, Johns Hopkins University; Space Operations Officer, United States Space Force

Ariadne Letrou (Lab manager and postgraduate researcher)

2021–2023

Now: PhD student, Princeton Psychology (PI: Ken Norman)

Kyle Andruczk (Yale undergraduate)

2022–2023

Now: Full stack software engineer

Ad-hoc reviewing

Nature Methods, Nature Human Behavior, Nature Computational Science, Journal of Neuroscience, Proceedings of the National Academy of Sciences (PNAS), Imaging Neuroscience, International Conference on Learning Representations (ICLR), Neural Information Processing Systems (NeurIPS), Cognitive Computational Neuroscience (CCN).

Other skills

Neuroimaging: rt-cloud (Real-time fMRI with cloud computing), MRI operator certified, MEG/EEG experienced. BrainIAK & PyMVPA Contributor, FSL, FreeSurfer, AFNI, Reproducible Neuroimaging.

Programming: Python, BASH, C, C++, C# for Unity, Java, MATLAB, R, HTML, Unity, PsychoPy, PsychToolbox, PyTorch, Keras, TensorFlow.

Languages: Spanish (fluent).

Miscellaneous: Competitive equestrian, runner, illustrator, freelance data scientist, general flora and fauna enthusiast.